Shuyang Liu

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Research Interest

My research lies at the intersection of Artificial Intelligence and Software Engineering, with a focus on understanding and optimizing software agents. I study how intelligent agents reason and act over long-horizon software engineering tasks, with the goal of building reliable, efficient, and adaptive agentic systems that can continuously improve their problem-solving capabilities.

EDUCATION

University of Illinois at Urbana-Champaign (UIUC) **Advisor: Reyhaneh Jabbarvand**
P.h.D in Computer Science *Aug. 2024 - Jun. 2029*

Huazhong University of Science and Technology **GPA: 3.93/4.00**
B.E. in Computer Science *Sept. 2020 - Jun. 2024*

PUBLICATION

Process-Centric Analysis of Agentic Software Systems
Shuyang Liu, Yang Chen, Rahul Krishna, Saurabh Sinha, Jatin Ganhotra, Reyhan Jabbarvand
Object-Oriented Programming, Systems, Languages & Applications (OOPSLA 2026)

From Plan to Action: How well do Agents Follow the Plan?
Shuyang Liu*, Saman Dehghan*, Jatin Ganhotra, Martin Hirzel, Reyhaneh Jabbarvand
41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)

Exploiting Code Symmetries for Learning Program Semantics
Kexin Pei, Weichen Li*, Qirui Jin*, **Shuyang Liu**, Scott Geng, Lorenzo Cavallaro, Junfeng Yang, Suman Jana
(**Spotlight Top-3.5%**) in 41st International Conference on Machine Learning (ICML 2024)

Graphectory Viewer: A Tool for Process-Centric Analysis of Agentic Software Trajectories
Charlie Jyu, **Shuyang Liu**, Reyhaneh Jabbarvand
41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026 Tools and Datasets Track)

Online Monitoring and Corrective Steering of Programming Agents
Shuyang Liu, Saman Dehghan, Jiyoung Kim, Jatin Ganhotra, Martin Hirzel, Reyhaneh Jabbarvand
Under Review

Narrowing the Complexity Gap in the Evaluation of Large Language Models
Yang Chen, **Shuyang Liu**, Reyhaneh Jabbarvand
Under Review

WORK EXPERIENCE

IBM Thomas J. Watson Research Center **Managers: Raju Pavuluri, Martin Hirzel**
Research Internship *May. 2025 - Aug. 2025*

COMMUNITY SERVICE

- Student Volunteer for SPLASH 2026.

SELECTED AWARDS

- ACM SIGSOFT CAPS Grant for ASE 2026.
- Outstanding Graduate of Class 2024, Huazhong University of Science and Technology.